

Vectorized Expectation-Maximization (EM) Algorithm for Gaussian Mixture Models (GMM)

Project Details

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 - **Course:** Bachelor of Computer Applications (BCA)
 - **Project:** Week 1 – Gaussian Mixture Model using Expectation-Maximization Algorithm
 - **Programming Language:** Base R
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Project Objective

This project implements the Gaussian Mixture Model (GMM) using the Expectation-Maximization (EM) Algorithm from scratch in Base R. The implementation does not use any machine learning or clustering libraries. The algorithm estimates the parameters of Gaussian distributions iteratively until convergence.

Software Used

- RStudio
 - Base R
 - Git
 - GitHub
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Project Structure

GMM-EM-Algorithm-Week1/

|— README.md

|— Week1_Report.pdf

|— synthetic_data.csv

|— gmm_em_algorithm.R

|— Output/

| ├── dataset.png
| ├── cluster1.png
| ├── cluster2.png
| ├── final_clusters.png
| └── log_likelihood.png

✦ Project Tasks

◆ Task 1 – Generate Dataset

- Generated a synthetic dataset with two Gaussian clusters
 - Total observations: 2,000,000 (20 lakh)
 - Saved the dataset as synthetic_data.csv
 - Plotted the generated dataset
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◆ Task 2 – Initialize GMM Parameters

- Number of clusters ($K = 2$)
 - Mixing coefficients
 - Mean vectors
 - Covariance matrices
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◆ Task 3 – Gaussian Density Function

- Implemented Multivariate Gaussian PDF using Base R
 - Computed probability density values
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◆ Task 4 – Expectation Step (E-Step)

- Calculated responsibility matrix
 - Ensured each row sums to 1
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◆ Task 5 – Maximization Step (M-Step)

Updated:

- Mixing coefficients
 - Mean vectors
 - Covariance matrices
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◆ Task 6 – EM Algorithm

- Repeated E-Step and M-Step
 - Computed Log-Likelihood
 - Stopped after convergence or 20 iterations
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◆ Task 7 – Visualization

Generated:

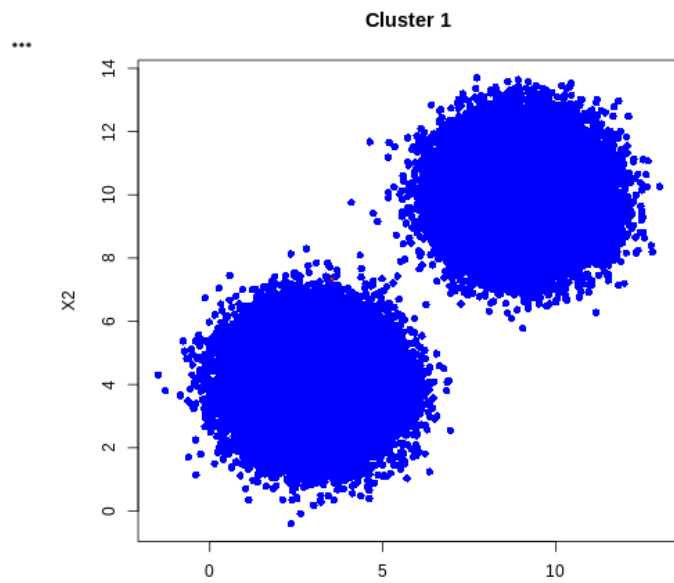
- Dataset plot
 - Final clusters
 - Log-likelihood graph
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How to Run

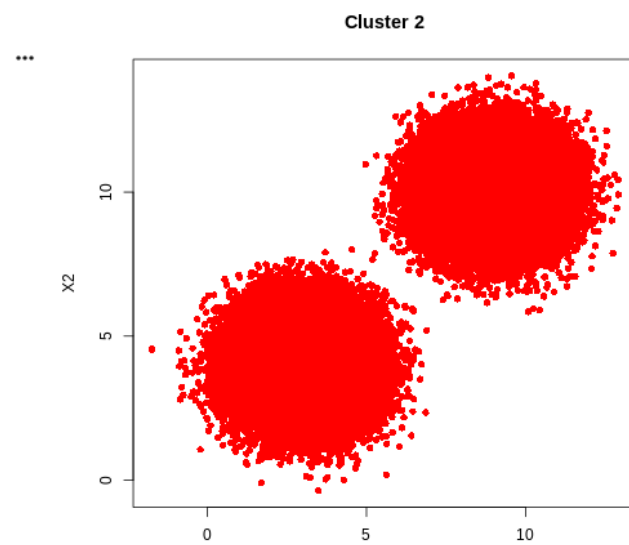
1. Open gmm_em_algorithm.R in RStudio
 2. Run the entire script
 3. Dataset and outputs will be generated automatically
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Output Screenshots

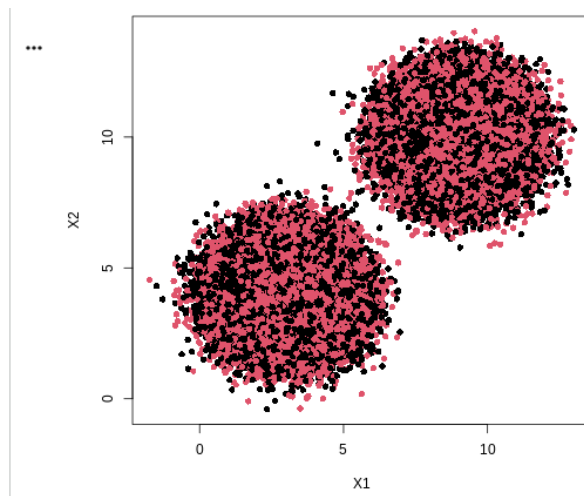
Cluster 1



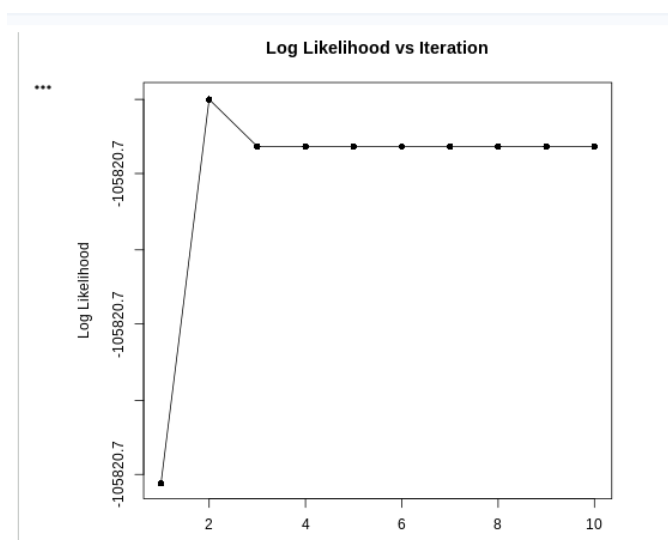
Cluster 2



Final Clusters



Log Likelihood



Learning Outcomes

- Understood Gaussian Mixture Models (GMM)
- Learned Expectation-Maximization (EM) Algorithm
- Implemented clustering from scratch
- Computed probabilities and likelihood
- Visualized clustering effectively
- Gained hands-on GitHub project experience